PHYLOGENETIC ANALYSIS OF THE HEMOGLOBIN BETA (HBB) GENE

1.0 INTRODUCTION:

Hemoglobin is an essential protein responsible for transporting oxygen from the lungs to tissues and carbon dioxide back to the lungs. In humans, adult hemoglobin is predominantly composed of two alpha (α) and two beta (β) globin subunits. The beta-globin subunit is encoded by the HBB (hemoglobin subunit beta) gene, making HBB an important gene and protein in human physiology and hematological disease.

The HBB gene belongs to the globin gene family, which has undergone evolutionary diversification while retaining important structural and functional characteristics. Due to hemoglobin's highly conserved physiological function, HBB homologues from different species provide a useful system for investigating sequence conservation, evolutionary divergence, and relationships among organisms.

Phylogenetic analysis provides a computational approach for reconstructing evolutionary relationships from biological sequence data. By comparing homologous protein sequences across species, conserved regions and sequence variations can be identified, and the evolutionary relatedness of the sequences can be represented in a phylogenetic tree.

The purpose of this project was to investigate the evolutionary relationships among HBB protein homologues using a comparative bioinformatics workflow. The human HBB protein sequence was used as the reference/query sequence. Homologous sequences were identified using BLASTp, aligned using ClustalW in MEGA 12, and subsequently used to construct a Neighbor-Joining phylogenetic tree. The resulting tree was further visualized and annotated using the Interactive Tree of Life (iTOL).

The project consisted of 31 protein sequences in total, comprising the human HBB query sequence and 30 homologous sequences identified through BLASTp.

2.0 METHODOLOGY:

2.1 Overall Workflow

The analysis followed a sequential computational workflow:

NCBI Protein Database

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Retrieval of human HBB protein sequence

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BLASTp homology search

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Selection of 30 homologous protein sequences

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Compilation of 31-sequence dataset

↓

Multiple sequence alignment using ClustalW in MEGA 12

↓

Neighbor-Joining phylogenetic tree construction

↓

Bootstrap analysis (1,000 replicates)

↓

Export of tree in Newick format

↓

iTOL visualization and annotation

2.2 Retrieval of the Human HBB Protein Sequence

Database used: NCBI Protein Database

The first stage involved retrieving the human hemoglobin subunit beta protein sequence from the **NCBI Protein Database**. The human HBB protein was identified using the accession **P68871.2.** This sequence served as the query/reference sequence for the subsequent BLASTp analysis. This is because a reliable reference sequence is required before homologous proteins can be identified. The human HBB sequence provided a standardized starting point for identifying related HBB proteins in other organisms.

The sequence was represented in FASTA format, which is a widely used format for storing biological nucleotide or protein sequences and allows it to be readily submitted to downstream bioinformatics programs. Other accompanying NCBI record files were retained as part of the project dataset.

The following files were downloaded:

Table 1.0: File formats for the NCBI Protein Retrieval

File	Format
Sequence	.fasta
Amino acid sequence and detailed annotation metadata	.gp
Feature Table	.txt

2.3 BLASTp Homology Search

Tool: BLASTp

BLASTp (Basic Local Alignment Search Tool for proteins) was used to identify proteins with sequences similar to the human HBB query sequence.

The human HBB protein was submitted as the query against the NCBI protein database. The search was used to identify homologous proteins based on sequence similarity. Particular attention was given to the following BLASTp parameters:

- ❖ Query coverage
- ❖ Percentage identity
- ❖ Sequence similarity
- ❖ E-value
- ❖ Alignment quality
- ❖ Subject accession and organism information

The analysis retrieved the top 30 homologous protein sequences. Together with the original human query sequence, these produced the final dataset of 31 protein sequences used for downstream analysis.

Table 2.0 File Formats for the BLASTp Homology Search

File	Format
Sequence Dump	.fasta
BLASTp Analysis	.gp
Alignment Text	.txt
Alignment Hit Table	.xlsx

Figure 1.0 BLASTp Analysis

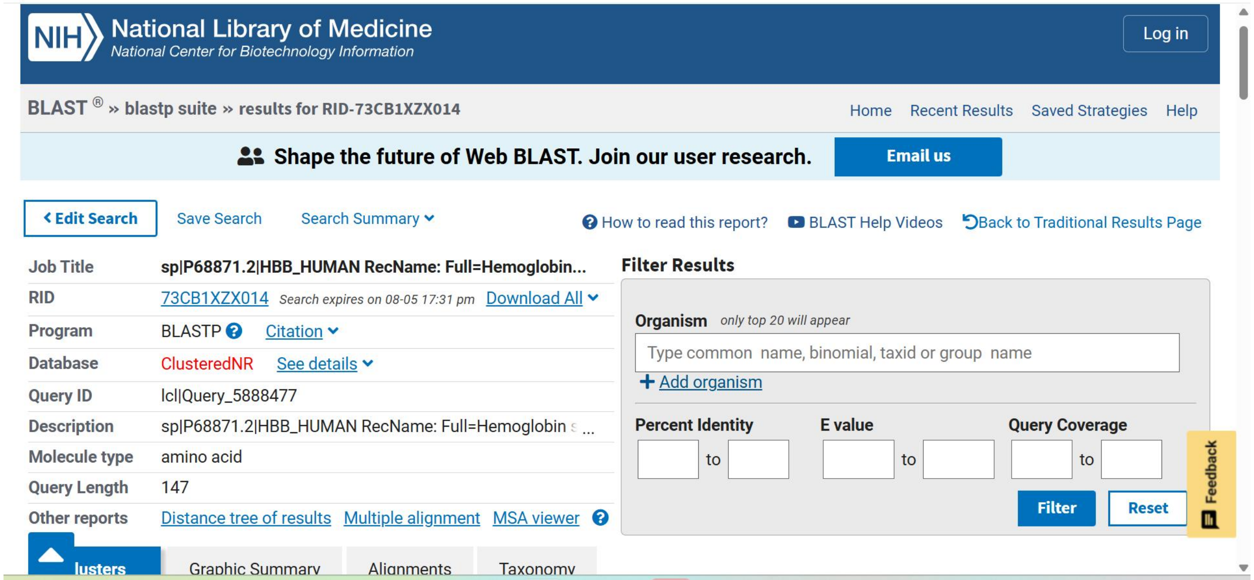
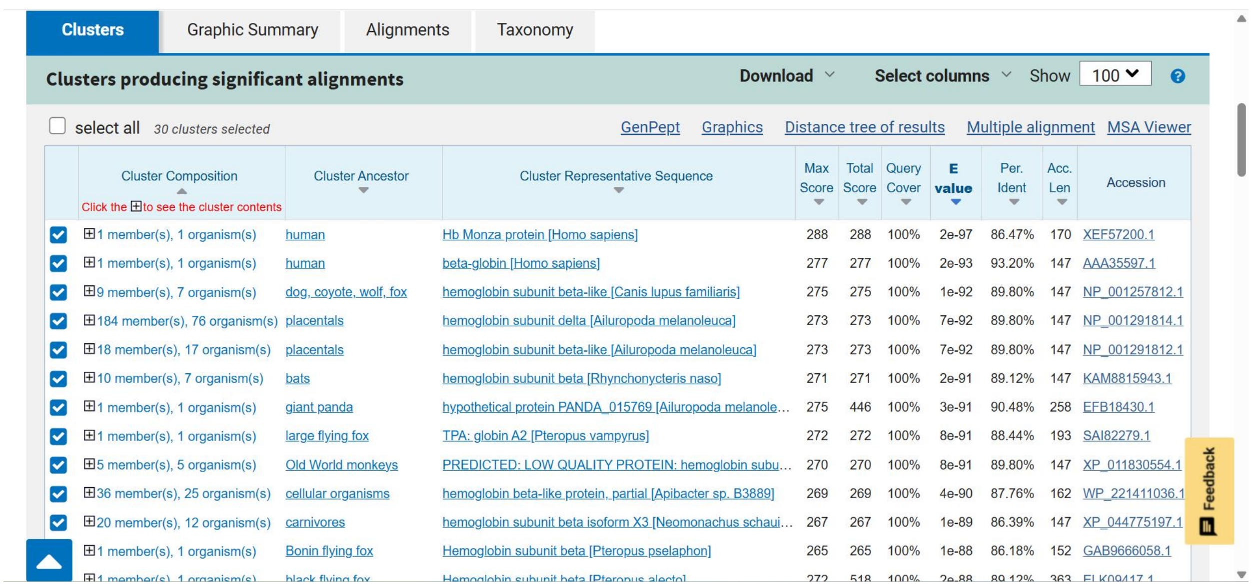


Figure 2.0 Top 30 BLAST hits (1)



Cluster Composition (Click the ⊞ to see the cluster contents)	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
1 member(s), 1 organism(s)	human	Hb Monza protein [Homo sapiens]	288	288	100%	2e-97	86.47%	170	XEF57200.1
1 member(s), 1 organism(s)	human	beta-globin [Homo sapiens]	277	277	100%	2e-93	93.20%	147	AAA35597.1
9 member(s), 7 organism(s)	dog, coyote, wolf, fox	hemoglobin subunit beta-like [Canis lupus familiaris]	275	275	100%	1e-92	89.80%	147	NP_001257812.1
184 member(s), 76 organism(s)	placentals	hemoglobin subunit delta [Ailuropoda melanoleuca]	273	273	100%	7e-92	89.80%	147	NP_001291814.1
18 member(s), 17 organism(s)	placentals	hemoglobin subunit beta-like [Ailuropoda melanoleuca]	273	273	100%	7e-92	89.80%	147	NP_001291812.1
10 member(s), 7 organism(s)	bats	hemoglobin subunit beta [Rhynchonycteris naso]	271	271	100%	2e-91	89.12%	147	KAM8815943.1
1 member(s), 1 organism(s)	giant panda	hypothetical protein PANDA_015769 [Ailuropoda melanole…	275	446	100%	3e-91	90.48%	258	EFB18430.1
1 member(s), 1 organism(s)	large flying fox	TPA: globin A2 [Pteropus vampyrus]	272	272	100%	8e-91	88.44%	193	SAI82279.1
5 member(s), 5 organism(s)	Old World monkeys	PREDICTED: LOW QUALITY PROTEIN: hemoglobin subu…	270	270	100%	8e-91	89.80%	147	XP_011830554.1
36 member(s), 25 organism(s)	cellular organisms	hemoglobin beta-like protein, partial [Apibacter sp. B3889]	269	269	100%	4e-90	87.76%	162	WP_221411036.1
20 member(s), 12 organism(s)	carnivores	hemoglobin subunit beta isoform X3 [Neomonachus schaui…	267	267	100%	1e-89	86.39%	147	XP_044775197.1
1 member(s), 1 organism(s)	Bonin flying fox	Hemoglobin subunit beta [Pteropus pselaphon]	265	265	100%	1e-88	86.18%	152	GAB9666058.1
1 member(s), 1 organism(s)	black flying fox	Hemoglobin subunit beta [Pteropus alecto]	272	518	100%	2e-88	89.12%	363	ELK09417.1

Figure 3.0 Top 30 BLAST hits (2)

⊞20 member(s), 12 organism(s)	carnivores	hemoglobin subunit beta isoform X3 [Neomonachus schaui…	267	267	100%	1e-89	86.39%	147	XP_044775197.1
⊞1 member(s), 1 organism(s)	Bonin flying fox	Hemoglobin subunit beta [Pteropus pselaphon]	265	265	100%	1e-88	86.18%	152	GAB9666058.1
⊞1 member(s), 1 organism(s)	black flying fox	Hemoglobin subunit beta [Pteropus alecto]	272	518	100%	2e-88	89.12%	363	ELK09417.1
⊞2 member(s), 2 organism(s)	placentals	RecName: Full=Hemoglobin subunit beta; AltName: Full=B…	264	264	99%	2e-88	86.30%	146	P11752.1
⊞1 member(s), 1 organism(s)	human	beta-globin [Homo sapiens]	263	263	87%	9e-88	100.00%	175	AAA35952.1
⊞1 member(s), 1 organism(s)	southern tamandua	hemoglobin subunit beta [Tamandua tetradactyla]	261	261	100%	2e-87	86.39%	147	XP_076969900.1
⊞1 member(s), 1 organism(s)	Yong Hoi Sen's woolly h…	hemoglobin subunit beta [Rhinolophus yonghoiseni]	263	263	100%	4e-87	85.71%	204	KAN3658820.1
⊞member(s), organism(s)	NA	hemoglobin beta [Tarsius sp.]	259	259	99%	1e-86	84.93%	146	765952B
⊞4 member(s), 4 organism(s)	placentals	hemoglobin subunit beta [Sorex araneus]	259	259	100%	1e-86	84.35%	147	XP_004618299.1
⊞8 member(s), 7 organism(s)	guinea pigs and others	hemoglobin subunit beta [Octodon degus]	258	258	100%	4e-86	83.67%	147	XP_004642055.1
⊞1 member(s), 1 organism(s)	raccoon	hemoglobin subunit beta [Procyon lotor]	258	258	100%	5e-86	81.63%	147	GAC2893078.1
⊞2 member(s), 1 organism(s)	northern gundi	RecName: Full=Hemoglobin subunit beta; AltName: Full=B…	257	257	99%	8e-86	84.25%	146	P20855.1
⊞6 member(s), 2 organism(s)	pigs	hemoglobin subunit beta [Sus scrofa]	257	257	100%	8e-86	85.03%	147	NP_001138313.1
⊞3 member(s), 2 organism(s)	hippopotamus	RecName: Full=Hemoglobin subunit beta; AltName: Full=B…	257	257	99%	9e-86	85.62%	146	P19016.1
⊞30 member(s), 19 organism(s)	carnivores	RecName: Full=Hemoglobin subunit beta; AltName: Full=B…	257	257	98%	1e-85	85.42%	146	P19646.1
⊞1 member(s), 1 organism(s)	lesser Egyptian jerboa	hemoglobin subunit beta [Jaculus jaculus]	256	256	100%	2e-85	82.31%	147	XP_004651012.1
⊞1 member(s), 1 organism(s)	southern two-toed sloth	hemoglobin subunit beta [Choloepus didactylus]	256	256	100%	3e-85	83.67%	147	XP_037696241.1
⊞3 member(s), 3 organism(s)	placentals	hemoglobin subunit epsilon isoform X2 [Acinonyx jubatus]	256	256	100%	3e-85	80.95%	147	XP_053059579.1
⊞4 member(s), 2 organism(s)	rodents	hemoglobin beta, partial [Callospermophilus lateralis]	255	255	100%	5e-85	81.63%	147	AGU26656.1
⊞4 member(s), 3 organism(s)	rodents	RecName: Full=Hemoglobin subunit beta; AltName: Full=B…	255	255	99%	6e-85	82.19%	146	P02090.1

Feedback

2.4 Multiple Sequence Alignment

Software: MEGA 12

Algorithm: ClustalW

The 31 protein sequences were imported into **MEGA 12** for multiple sequence alignment. This was performed to facilitate phylogenetic analysis by aligning homologous residues at corresponding positions across all sequences. This makes it possible to identify positions that have remained conserved throughout evolution as well as positions where amino acid substitutions or other sequence differences have occurred. The **ClustalW** algorithm was used to perform the alignment, and the completed alignment was subsequently used as the input for phylogenetic tree construction.

The multiple sequence alignment served two major purposes:

- ❖ **Identification of conserved regions:** Amino acid positions that remain similar across many or all sequences may represent regions that are important for the structure or function of HBB.
- ❖ **Identification of evolutionary variation:**
  Amino acid substitutions among species provide evidence of sequence divergence and contribute information used to reconstruct evolutionary relationships.

Table 3.0 File Formats for the Multiple Sequence Alignment

File	Format
Input File	.fas
Raw Sequence Dump	.fasta
MEGA Alignment File	.meg
MEGA 12 File	.mas

2.5 Phylogenetic Tree Construction

Software: MEGA 12

Method: Neighbour-Joining

The aligned HBB protein sequences were used to construct a phylogenetic tree using the **Neighbor-Joining (NJ)** method in MEGA 12. In this analysis, sequences with greater similarity were expected to occur closer together on the tree, while sequences with greater evolutionary divergence were expected to occur farther apart.

Bootstrap Analysis:

To evaluate the reliability of the inferred tree topology, **1,000 bootstrap replicates** were performed.

Bootstrap analysis involves repeatedly resampling alignment positions and reconstructing trees from the resampled datasets. The percentage of bootstrap replicates that support a given branch is expressed as its bootstrap value.

Higher bootstrap values generally indicate stronger support for the corresponding grouping, while lower values indicate greater uncertainty.

The resulting phylogenetic tree was exported in Newick (.nwk) format for subsequent visualization.

Table 4.0 File Formats for the Phylogenetic Tree Construction

File	Format
Newick Tree	.nwk
MEGA-generated Tree	.png
MEGA-generated Tree	.svg

2.6 Phylogenetic Tree Visualization and Annotation

Tool: Interactive Tree of Life (iTOL)

The Newick-format tree generated in MEGA 12 was imported into the **Interactive Tree of Life (iTOL)** for visualization and annotation. Although MEGA 12 was used to generate the phylogenetic tree, iTOL provided greater flexibility for presenting the evolutionary relationships in a visually interpretable form.

During this stage:

- ❖ Sequence/species labels were formatted for readability.
- ❖ The tree layout was adjusted to a radial representation.
- ❖ Branch presentation was refined.
- ❖ Labels were organized to make relationships easier to interpret.
- ❖ The final tree was exported as a high-quality graphical representation.

The refined visualization made it easier to identify clusters of related HBB homologs and compare their evolutionary relationships across the sampled vertebrate lineages.

Table 5.0 File Formats for the Phylogenetic Tree Visualization and Annotation

File	**Format**
Original Rectangular Tree	.svg
Refined Radial Tree	.png
Refined Radial Tree	.svg

3.0 TOOLS AND DATABASES USED:

Table 6.0 Tools and Databases used during this Project

Tool/Database	**Role in the Project**
NCBI Protein Database	Retrieval of the human HBB reference protein sequence
BLASTp	Identification of homologous HBB protein sequences
MEGA 12	Multiple sequence alignment and phylogenetic analysis
ClustalW	Multiple sequence alignment algorithm implemented in MEGA 12
Neighbor-Joining	Phylogenetic tree construction method

Bootstrap analysis	Assessment of support for inferred phylogenetic relationships
iTOL	Phylogenetic tree visualization and annotation
FASTA	Storage and exchange of protein sequence data
Newick	Representation and transfer of the phylogenetic tree

4.0 RESULTS:

The analysis produced a comparative dataset of 31 HBB protein sequences, comprising the human HBB query sequence and 30 homologues identified using BLASTp.

The BLASTp analysis demonstrated that the query protein had detectable homologues among the selected vertebrate proteins. The identified homologous sequences were then subjected to multiple sequence alignment using ClustalW to assess conserved and variable amino acid residues across the dataset.

The resulting alignment provided the basis for phylogenetic inference. A Neighbor-Joining tree was constructed in MEGA 12 and evaluated using 1,000 bootstrap replicates. The resulting tree was exported in Newick format and subsequently refined using iTOL.

The phylogenetic visualization demonstrated that the HBB homologues did not occur randomly within the tree. Instead, sequences formed groups based on their sequence similarities and evolutionary relationships. Closely related sequences were positioned within the same or neighboring clades, whereas more divergent sequences occurred on more distant branches.

The observed clustering reflects the evolutionary conservation of the HBB protein among vertebrates, while highlighting sequence differences that have emerged as species diverged over time.

Importantly, the phylogenetic tree should be interpreted as a representation of relationships inferred from the HBB protein sequences included in this project, rather than as an absolute species tree. Differences between a gene/protein tree and the actual species evolutionary history can arise from processes such as gene duplication, gene loss, incomplete lineage sorting, or differences in evolutionary rates.

5.0 DISCUSSION

The comparative analysis of HBB homologues demonstrates how sequence-based bioinformatics can be used to investigate molecular evolution without requiring laboratory-based experimental procedures.

The first major observation was the ability to identify homologous HBB proteins using BLASTp. The presence of homologous sequences across multiple vertebrate species is consistent with the evolutionary conservation of hemoglobin and its essential role in oxygen transport.

The multiple sequence alignment enabled the assessment of amino acid conservation across the analyzed sequences. Highly conserved positions across the dataset suggest that certain regions of the HBB protein have been maintained by evolutionary pressure. Such conservation is biologically meaningful because changes in functionally important residues may negatively affect protein structure or activity.

Conversely, amino acid differences observed among homologues demonstrate that the HBB protein has also undergone evolutionary divergence. These substitutions contribute to the phylogenetic signal used to distinguish closely and distantly related sequences.

The Neighbor-Joining tree provided a graphical representation of these sequence relationships. The proximity of sequences within the phylogenetic tree reflects their degree of similarity, with greater separation indicating higher evolutionary divergence.

Bootstrap analysis enhanced the reliability of the phylogenetic analysis by indicating how consistently individual branches were recovered across replicate datasets. Nevertheless, bootstrap values should be considered alongside sequence quality, taxon sampling, alignment accuracy, and the phylogenetic method employed, rather than being regarded as definitive evidence of evolutionary relationships.

The use of iTOL enhanced interpretation by converting the Newick tree into a clearer and more presentable visualization. This was particularly useful for identifying clusters and communicating the evolutionary relationships among the HBB homologues.

Overall, the project demonstrates a complete computational phylogenetic workflow beginning with sequence retrieval and ending with an interpretable evolutionary tree.

6.0 CONCLUSION:

This project successfully applied a complete bioinformatics workflow to investigate the evolutionary relationships among hemoglobin subunit beta (HBB) protein homologues across vertebrate lineages.

The human HBB protein sequence, identified by accession **P68871.2**, was retrieved from the NCBI Protein Database and used as the reference sequence. BLASTp was subsequently used to identify 30 homologous protein sequences, resulting in a final dataset of 31 sequences including the query sequence.

ClustalW, implemented in MEGA 12, was used to generate a multiple sequence alignment, facilitating the assessment of amino acid conservation and variation among the sequences. The resulting alignment was used to construct a Neighbor-Joining phylogenetic tree, with 1,000 bootstrap replicates used to evaluate the reliability of the inferred relationships.

The tree was subsequently exported in Newick format and visualized and refined using iTOL. The resulting phylogenetic representation demonstrated evolutionary relationships among the sampled HBB homologues, with closely related sequences clustering together based on their sequence similarities.

Overall, the project highlights the value of bioinformatics in studying molecular evolution and demonstrates how publicly available sequence databases and computational tools can be integrated to investigate gene/protein conservation and evolutionary divergence.

The workflow also provided practical experience in sequence retrieval, protein homology searching, multiple sequence alignment, phylogenetic inference, bootstrap analysis, Newick tree handling and scientific visualization. These skills are broadly applicable to comparative genomics, molecular evolution, structural biology and biomedical research.